

HDOC – Day_4

Problem Statement :

Correct the given C program, algorithm, and flowchart to perform multiplication of two integers without using the multiplication (*) operator. Implement the solution using repeated addition and ensure that it correctly handles positive numbers, negative numbers, and zero.

- **Objective:**

To implement multiplication of two integers using repeated addition instead of the multiplication (*) operator and verify the program using different test cases.

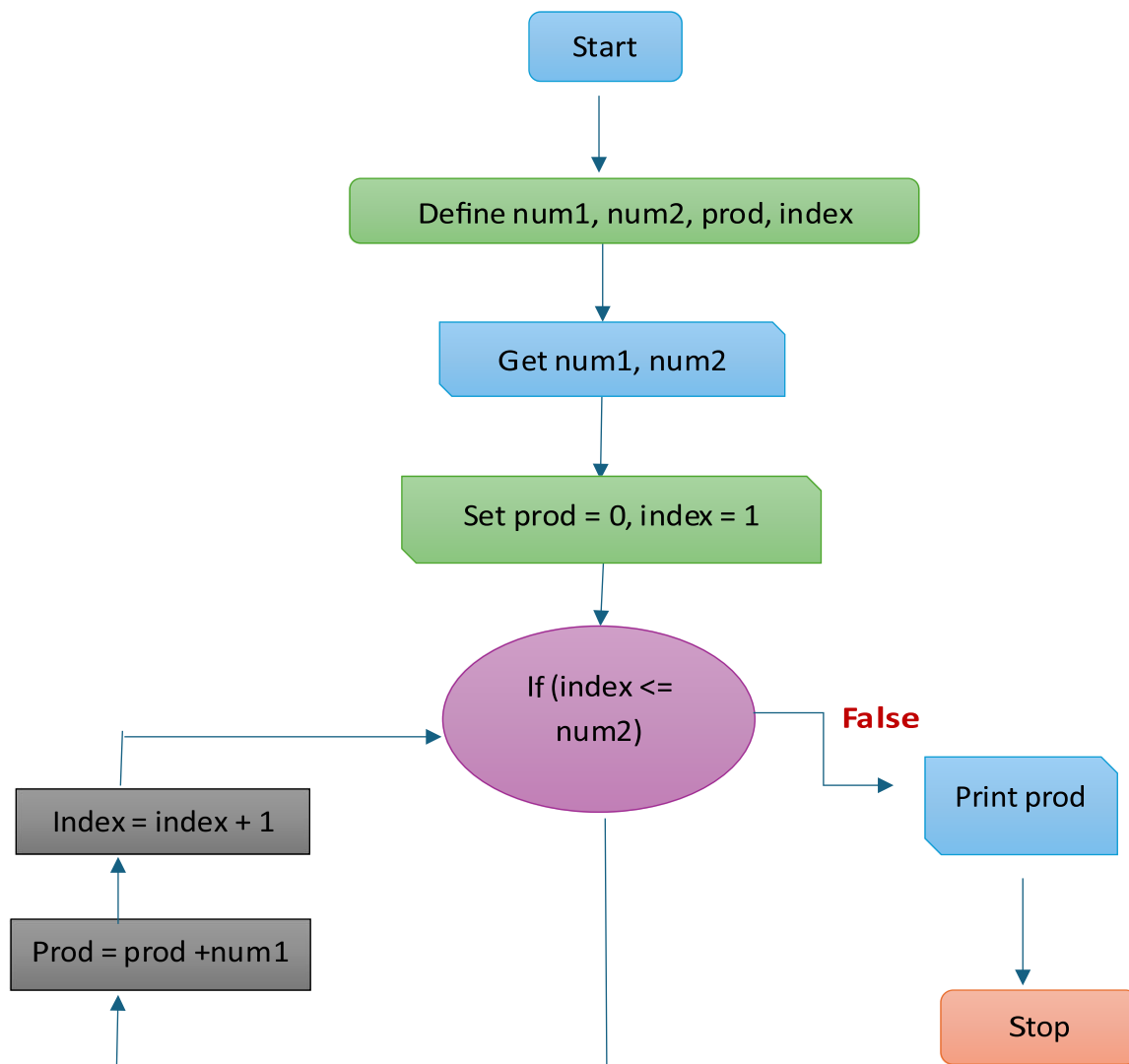
- **Given Algorithm:**

- 1) Define num1, num2, prod, index as integers.
- 2) Get num1, num2.
- 3) Set prod = 0, index = 1
- 4) Repeat steps 4.1 and 4.2 until (index <= num2)
 - 4.1 prod = prod + num1
 - 4.2 index = index + 1
- 5) Print prod
- 6) Stop

- **The mistakes in the given algorithm are :**

1. It works only for positive values of num2.
2. It does not work for negative numbers.
3. It does not correctly handle multiplication if one of the inputs is zero.
4. It does not determine the sign of the result.

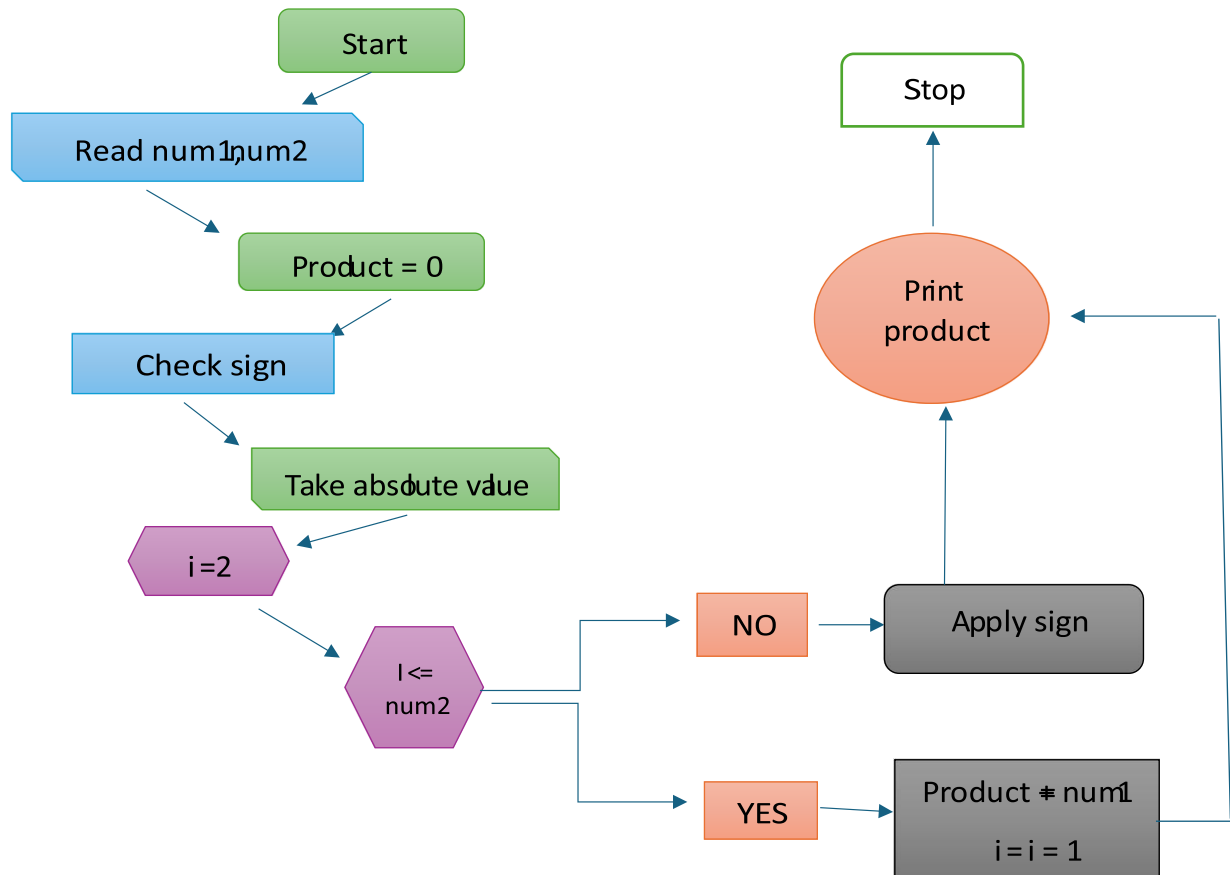
- **Flowchart**



The correct algorithm :

1. Start.
2. Read two integers num1 and num2.
3. Initialize product = 0.
4. Determine whether the result should be negative.
5. Convert num1 and num2 to their absolute values.
6. Repeat num2 times:
 7. $\text{product} = \text{product} + \text{num1}$
8. If the result should be negative, make $\text{product} = -\text{product}$.
9. Display the product.
10. Stop.

The correct flowchart:



CONCLUSION

This activity depicted that we can get the product of two numbers without using the multiplication operator, we can use repetitive addition instead of the multiply operator to get the desired output. The program has given correct outputs for positive numbers, negative numbers as well as for zero.